MEASUREMENTS AND UNITS

Introduction

Measurement is an important aspect of Physics. Physics deals with quantities such as length, mass, time, temperature, force, speed, and energy. To study these quantities accurately, standard units are required.

Measurement allows scientists and students to:

- Compare quantities
- Carry out experiments
- Record observations accurately
- Communicate scientific results clearly

Without standard measurements, scientific work would be confusing and unreliable.

Meaning of Measurement

Measurement is the process of determining the size, amount, or magnitude of a physical quantity by comparing it with a standard unit.

Example:

- Measuring the length of a table using a metre rule
- Measuring the mass of an object using a balance
- Measuring time using a stopwatch

Physical Quantities

A physical quantity is any quantity that can be measured and expressed with a number and a unit.

Examples:

- Length
- Mass
- Time

- Temperature
- Volume
- Force

Every physical quantity has:

1. Numerical value
2. Unit

Example:
5 metres

5 = numerical value
metre = unit

Fundamental (Basic) Quantities

Fundamental quantities are quantities that cannot be derived from other quantities.

There are seven SI fundamental quantities.

Fundamental Quantity	SI Unit	Symbol
Length	Metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	Kelvin	K
Amount of substance	Mole	mol
Luminous intensity	candela	cd

Derived Quantities

Derived quantities are quantities obtained from combinations of fundamental quantities.

Examples:

Derived Quantity	**Formula**	**SI Unit**
Area	length × breadth	m^2
Volume	length × breadth × height	m^3
Speed	$\frac{distance}{time\ taken}$	m/s
Density	$\frac{mass}{volume}$	kg/m^3
Force	mass × acceleration	N
Pressure	$\frac{force}{area}$	Pa

Units

A unit is a standard quantity used for measurement.

Example:

- metre for length
- kilogram for mass
- second for time

SI Units

SI means International System of Units.

It is the standard system of measurement used worldwide.

Advantages of SI Units:

1. Universal acceptance
2. Simplicity
3. Accuracy
4. Easy conversion

Prefixes Used in Measurements

Prefixes are used to express very large or very small quantities.

Prefix	Symbol	Value
kilo	K	1000
hecto	H	100
deca	da	10
deci	D	0.1
centi	C	0.01
milli	m	0.001
micro	µ	0.000001

Examples:

- 1 kilometre = 1000 metres
- 1 centimetre = 0.01 metre
- 1 milligram = 0.001 gram

Measuring Instruments

1. Metre Rule

Used for measuring length.

Features:

- Usually 1 metre long
- Graduated in centimetres and millimetres

Uses:

- Measuring length of objects
- Measuring distances

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2. Vernier Calipers

Used for measuring:

- Internal diameter
- External diameter
- Depth of objects

Accuracy:

- More accurate than metre rule

Least count:
Usually 0.01 cm

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3. Micrometer Screw Gauge

Used for measuring very small thicknesses and diameters.

Examples:

- Diameter of a wire
- Thickness of paper

Accuracy:
Very accurate

Least count:
Usually 0.01 mm

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4. Stopwatch

Used for measuring time intervals.

Examples:

- Time taken for a race
- Oscillation of a pendulum

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5. Beam Balance

Used for measuring mass.

Errors in Measurement

No measurement is perfectly accurate. Errors may occur during measurement.

Types of Errors:

1. Systematic Errors

These occur due to faults in instruments or wrong methods.

Examples:

- Faulty metre rule
- Incorrect calibration

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2. Random Errors

These occur due to changes in observation conditions.

Examples:

- Human reaction time
- Fluctuations in readings

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3. Personal Errors

Caused by the observer.

Examples:

- Wrong reading
- Wrong recording

Parallax Error

Parallax error occurs when the eye is not placed directly in line with the scale reading.

To avoid parallax error:

- Read scales at eye level

Accuracy

Accuracy refers to how close a measured value is to the true value.

Example:
If the actual length is 10 cm and the measured value is 10.01 cm, the measurement is accurate.

Precision

Precision refers to how close repeated measurements are to one another.

Example:
10.2 cm, 10.2 cm, 10.2 cm

The measurements are precise.

Significant Figures

Significant figures are meaningful digits in a measurement.

Rules:

1. All non-zero digits are significant.
2. Zeros between non-zero digits are significant.
3. Leading zeros are not significant.
4. Trailing zeros after decimals are significant.

Examples:

- 45.6 has 3 significant figures
- 0.0045 has 2 significant figures
- 3.200 has 4 significant figures

Decimal Places

Decimal places refer to the number of digits after the decimal point.

Examples:

- 5.32 has 2 decimal places
- 7.456 has 3 decimal places

Scientific Notation (Standard Form)

Scientific notation is used for writing very large or very small numbers.

General form:

$A \times 10^n$

Where:

- A is a number between 1 and 10
- n is an integer

Examples:

- $3000 = 3 \times 10^3$
- $0.00045 = 4.5 \times 10^{-4}$

Dimension of Physical Quantities

Dimensions show how physical quantities relate to fundamental quantities.

Examples:

Quantity	Dimension
Velocity	LT^{-1}
Force	MLT^{-2}
Density	ML^{-3}

Where:

- M = Mass

- L = Length
- T = Time

Scalars and Vectors

Scalar Quantities

Quantities with magnitude only.

Examples:

- Mass
- Time
- Temperature
- Distance

Vector Quantities

Quantities with magnitude and direction.

Examples:

- Force
- Velocity
- Acceleration
- Displacement

Worked Examples

Example 1

Convert 5 km to metres.

1 km = 1000 m

5 km = 5 × 1000

= 5000 m

Example 2

Convert 250 cm to metres.

100 cm = 1 m

250 cm = 250 ÷ 100

= 2.5 m

Example 3

A student measures the mass of an object as 2.50 kg. State the number of significant figures.

2.50 has 3 significant figures.

Example 4

Express 0.00072 in standard form.

$0.00072 = 7.2 \times 10^{-4}$

Example 5

Calculate speed if a car travels 100 m in 20 s.

Speed = distance ÷ time

= 100 ÷ 20

= 5 m/s

Importance of Measurements

1. Ensures accuracy in experiments
2. Helps in engineering and construction
3. Useful in medicine
4. Important in transportation

5. Necessary in scientific research

Practice Questions

1. Define measurement.
2. State four fundamental quantities.
3. Mention three measuring instruments and their uses.
4. Convert:
   a. 8 km to metres
   b. 450 cm to metres
5. Define:
   a. Accuracy
   b. Precision
6. State three types of measurement errors.
7. What is parallax error?
8. Express the following in standard form:
   a. 45000
   b. 0.00056
9. Differentiate between scalar and vector quantities.
10. Calculate the speed of a car that travels 200 m in 25 s.

Conclusion

Measurement and units form the foundation of Physics. Every physical quantity must be measured using standard units to ensure accuracy and consistency. Understanding measuring instruments, SI units, errors, significant figures, and scientific notation helps students perform experiments correctly and solve Physics problems effectively.